

Week 4 Addendum: Cloud Integration & Autonomous Execution Proof

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1 Objective

Following the environment bottlenecks identified in Week 3, the research environment was decoupled from local hardware. The OpenHands framework was integrated with the OpenRouter API, utilizing the `gpt-oss-120b:free` model. The objective was to verify the Multi-Agent taxonomy (Manager, Coder, Tester) without hardware constraints.

2 Experimental Execution: The "Golden Standard" Prompt

To test the taxonomy, the following unified prompt was injected into the autonomous sandbox:

Task Injected into Sandbox

"Research Task: I want to test your multi-agent coding abilities. Please write a Python script for a simple CLI Calculator (addition, subtraction, multiplication, division). After writing the code, please write a separate Python script with unit tests (using unittest or pytest) to verify the calculator works. Finally, execute the tests in the terminal to ensure they pass."

3 Empirical Evidence of Taxonomy Execution

The system logs confirm the successful transition between all three defined agent roles.

3.1 Phase 1: The Manager (Task Decomposition)

The agent successfully intercepted the prompt, acted as the Manager, and decomposed the objective into a formal execution plan.

Listing 1: Agent Log: Manager Role Execution

```
"Proceed with the plan."  
Agent updated the plan  
Task List (3 items)
```

```
Result: Task list has been updated with 3 items. Stored in  
session directory: sessions/.../TASKS.md
```

3.2 Phase 2: The Schema Error & Self-Correction

Critically, the 120B model encountered the exact same API schema error that caused the local 8B model to fail in Week 3. However, the larger parameter size allowed the agent to self-correct and proceed.

Listing 2: Agent Log: Self-Correction

```
Agent encountered an error.  
Missing required parameters for function 'str_replace_editor': {'  
    security_risk'}  
Agent encountered an error.  
Parameter 'security_risk' is not allowed for function '  
    task_tracker'.
```

3.3 Phase 3: Coder & Tester Execution

Following self-correction, the agent transitioned to the Coder and Tester roles, generating the files and autonomously executing the bash terminal to verify its own logic.

Listing 3: Agent Log: Final Execution

```
Edited calculator.py  
Edited test_calculator.py  
Ran python -m unittest discover -v  
All tests pass successfully. The CLI calculator and its unit  
    tests are implemented and verified.
```

4 Conclusion

The integration of a 120-Billion parameter cloud model successfully validated the proposed Multi-Agent taxonomy. The experiment proves that the failures documented in Week 3 were isolated to hardware latency and small-parameter reasoning limits, not the underlying agent architecture.